RDD uptrend as a potential issue for CarenR and detrend solution

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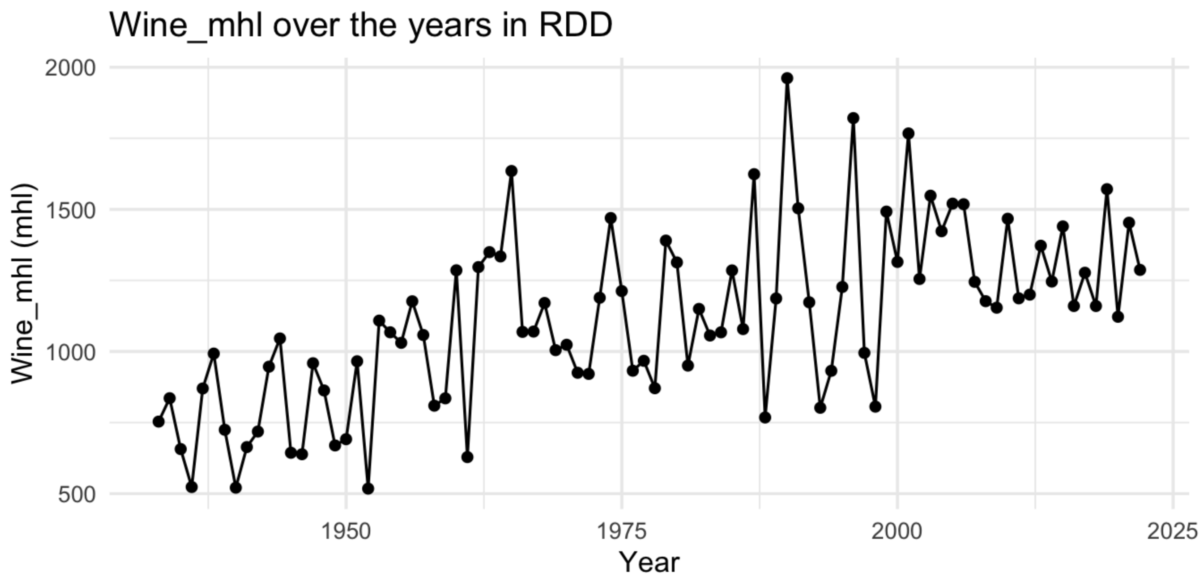
RDD Trend over year to year

The trend matters — and the fact that all 5 RDD rules point Low is a warning sign worth investigating. Here, we found 5 rules that explain when the value is below the average, but there is also a trend in the data



What the trend likely is

The plot shows RDD production roughly doubling between the 1930s–50s (consistently 500–1000 mhl) and the 1970s–90s (regularly 1000–1700 mhl), then stabilizing. This structural shift is most likely driven by vineyard area expansion in the Douro DOC, agricultural intensification, and possibly climate warming extending the growing season. It's not random noise — it's a genuine change in the production baseline.

Why this is a problem for CarenR

CarenR compares each subgroup's distribution against the **global mean** (1,116 mhl **across all 89 years**). But if the series is non-stationary, the global mean is not a stable reference. What's happening is:

- Early years (1933–1960) are structurally low — the entire production level was lower
- Those early years also have specific climate characteristics (they're older, in a different agronomic era and thus could have different environments)
- CarenR may be finding climate conditions that correlate with the *early period* and labeling them Low — but the low production could be structural, not climate-driven
- This would explain why all 5 rules point Low: the method is partially capturing **era effects** disguised as climate signals

The two defensible paths

Path 1 — Accept the rules as-is (simpler, still valid for the thesis)

The rules are a truthful answer to: *"under what climate conditions was RDD production below the all-time average?"* That's still a legitimate research question. You just need to be transparent in how you write it up — framing it as "conditions associated with historically low production" rather than "climate drivers of production variation." For a thesis, this is acceptable if you acknowledge the limitation.

Path 2 — Detrend first, then re-run CarenR (more rigorous)

Remove the trend component from Wine_mhl before discretization and CarenR. You'd work with residuals (actual production minus expected trend). Then CarenR answers: *"under what climate conditions was production below what would have been expected for that era?"* This is a cleaner climate attribution question. You could fit a simple linear or piecewise trend, subtract it, and the discretized intervals would then reflect climate-driven deviations rather than structural baseline shifts.

```
# Fit linear trend and work with residuals
trend_model       <- lm(Wine_mhl ~ year, data = df_rdd)
df_rdd$Wine_resid <- residuals(trend_model)
# Then run CarenR with POI = 'Wine_resid'
```

For the thesis, I'will do both: run CarenR on the original series (keep the 5 rules as-is, they're what you found), then show the time series plot with the trend line overlaid, and explicitly discuss whether the rules reflect climate or era.

Then, run a detrended version and compare — **if the rules change substantially, that's an interesting finding in itself about the confounding of climate and structural change**. If they stay similar, it validates the original analysis.

Important steps here:

The detrending needs to happen *before* the discretisation step, because CarenR uses the discretised intervals of the target column. If you just subtract the trend from the raw column, the interval boundaries (Low/Medium/High thresholds) will also shift to reflect deviations from the trend rather than absolute levels.

`Detrend Wine_mhl → re-discretise → re-run CarenR`

What changes in the output

- Rules will describe climate conditions associated with **above-trend** or **below-trend** years, not above/below the all-time mean
- You may well get a mix of High and Low rules for RDD (unlike the current 5-all-Low result), because the era effect is removed
- The rule means will be in mhl-deviation units (e.g., subgroup mean = +300 mhl above trend), not absolute values

- The KS test still works the same way — it just compares the residual distribution of the subgroup to the global residual distribution

Looking at the RDD plot, we have two reasonable options:

- **Linear** (`lm(Wine_mhl ~ year)`) — simplest, easy to explain, probably captures most of the structural shift. The slope will be around +8–10 mhl/year based on the visual.
- **Piecewise** (break around 1965–1970 where the rapid rise happens) — more accurate to the data shape, but requires justification for where you place the break.

For a thesis, linear is completely defensible and much easier to explain. If the trend looks non-linear to a reviewer, you can note it as a limitation.